

Problem Statement

- A thorough understanding of people's travel behavior, particularly in learning where people travel to, can yield numerous benefits
- Travel behavior data is typically collected through travel surveys, which are time-consuming and require substantial labor resources
- The promising social media data (SMD) can provide a large amount of travel behavior data, especially trip destination information, at a limited cost
- Previous social media data-based travel behavior analysis mainly utilized geotagged posts for constructing datasets, which may introduce incomplete dataset issue
- Natural language processing techniques, such as named-entity-matching (NEM), and named-entity-recognition, can be utilized to extract location entities from the contents of SMD
- Compared to named-entity-recognition, NEM is easy to be implemented and is more suitable to extract location information from informal social media posts

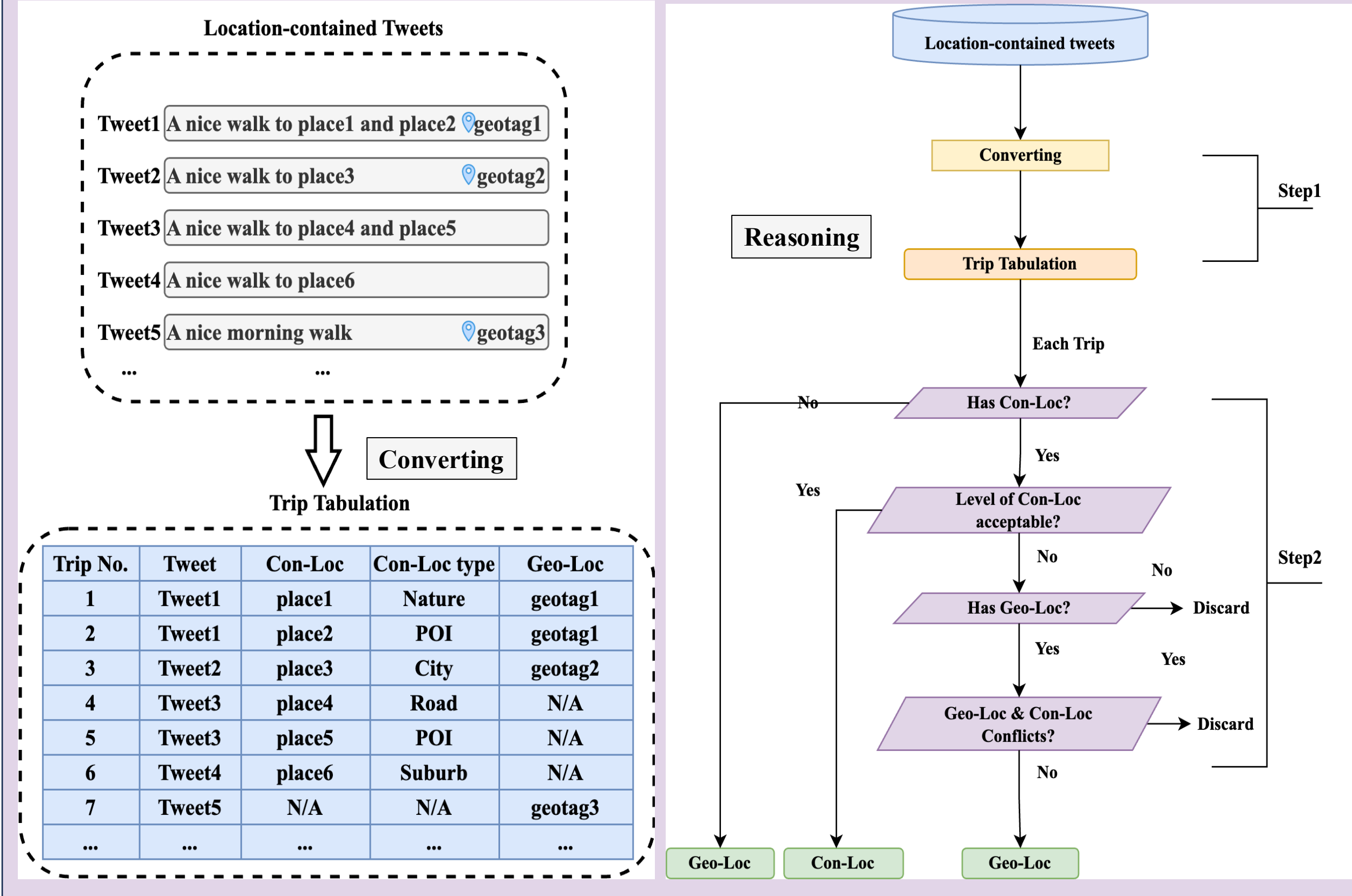
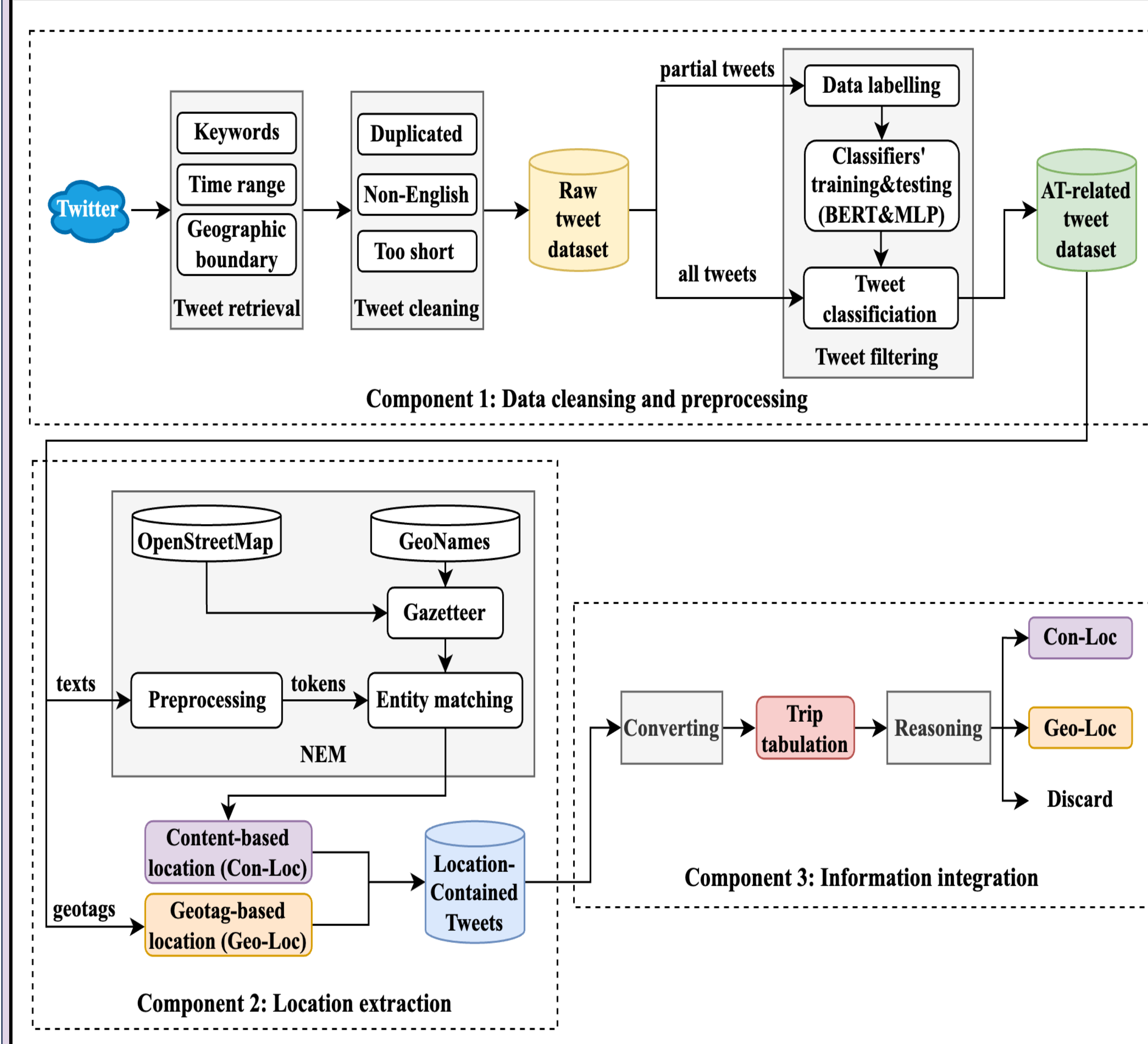
Research Aim

- This study mitigates the incomplete dataset issue by proposing a novel framework that supplements geotagged locations with content-based locations from social media data for travel behavior analysis
- The framework is showcased in Greater Melbourne, where Twitter data is utilized to investigate the travel behavior variation of active transporters due to the COVID-19

Methodology - Overview

- The specific implementation steps of the novel framework are introduced as follows:
- First, topic-related SMD is selected utilizing a Bidirectional Encoder Representations from Transformers (BERT) model-based classifier
- Second, content-based locations are extracted from the contexts of SMD using the NEM method, and geotag-based locations are identified from geotags
- Third, the two types of locations were integrated through a reasoning scheme

Methodology – Framework



Case Study

- A case study was conducted to evaluate the efficacy of the proposed framework.
- The case study utilized Twitter data (tweets) to investigate alterations in people's travel behavior on active transport in Greater Melbourne, Australia, during the pre-COVID-19 period (Pre-COV: 2018 and 2019) and the pandemic period (COV: 2020 and 2021).

Table 1. Keywords in tweet retrieval

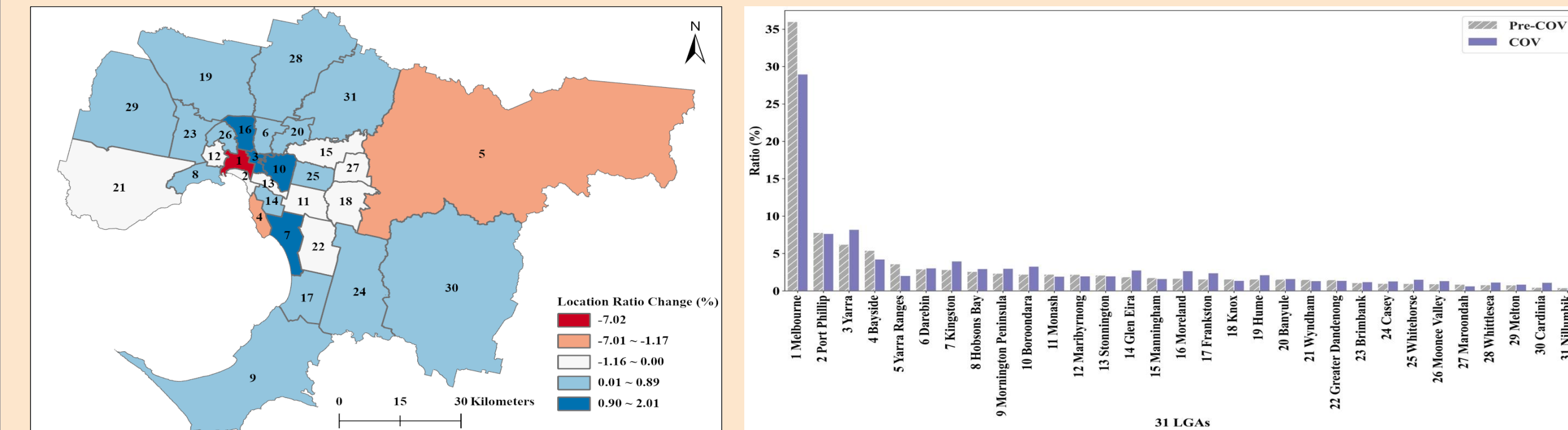
Classes	Keywords
Walking	'walk', 'walked', 'walker', 'walking', 'jog', 'jogger', 'jogging', 'pedestrian', 'run', 'running', 'ran', 'runner'
Cycling	'ride', 'rider', 'bike', 'biker', 'bicycle', 'bicyclist', 'cycling', 'cyclist'

Results & Discussion

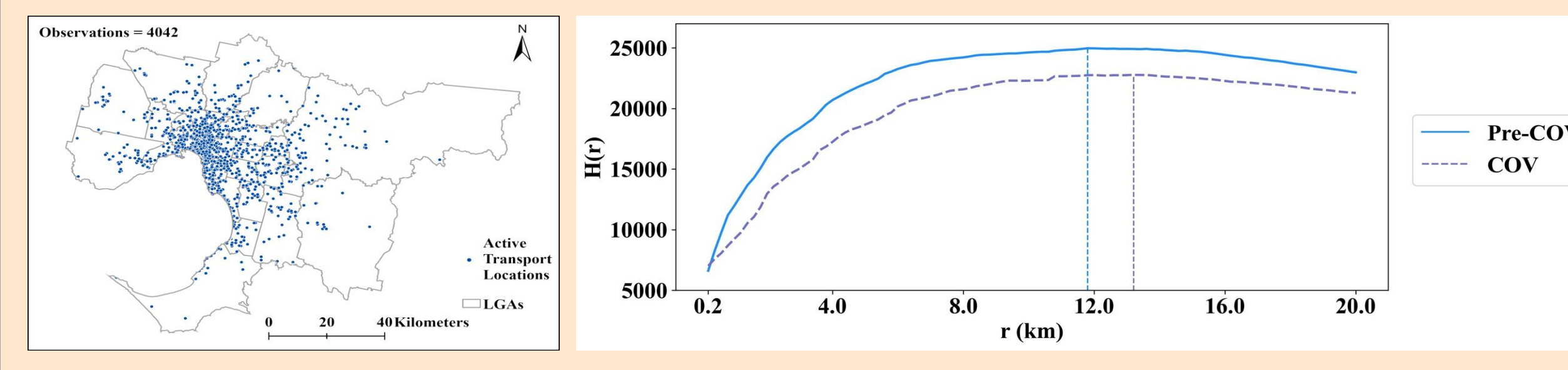
In the case study, 5330 location-contained tweets were identified. Compared to only collecting geotagged tweets, the framework increased the extraction of location-contained Twitter data by 33.70%

The number of active transport-related trips decreased in all cities (LGAs) within Greater Melbourne during the COV period, while the extent of reduction differed in the city. These variations were associated with cities' sociodemographic and built environment conditions

(Pre-COV: Pre-COVID-19 period, 2018&2019; COV: COVID-19 period, 2020&2021)

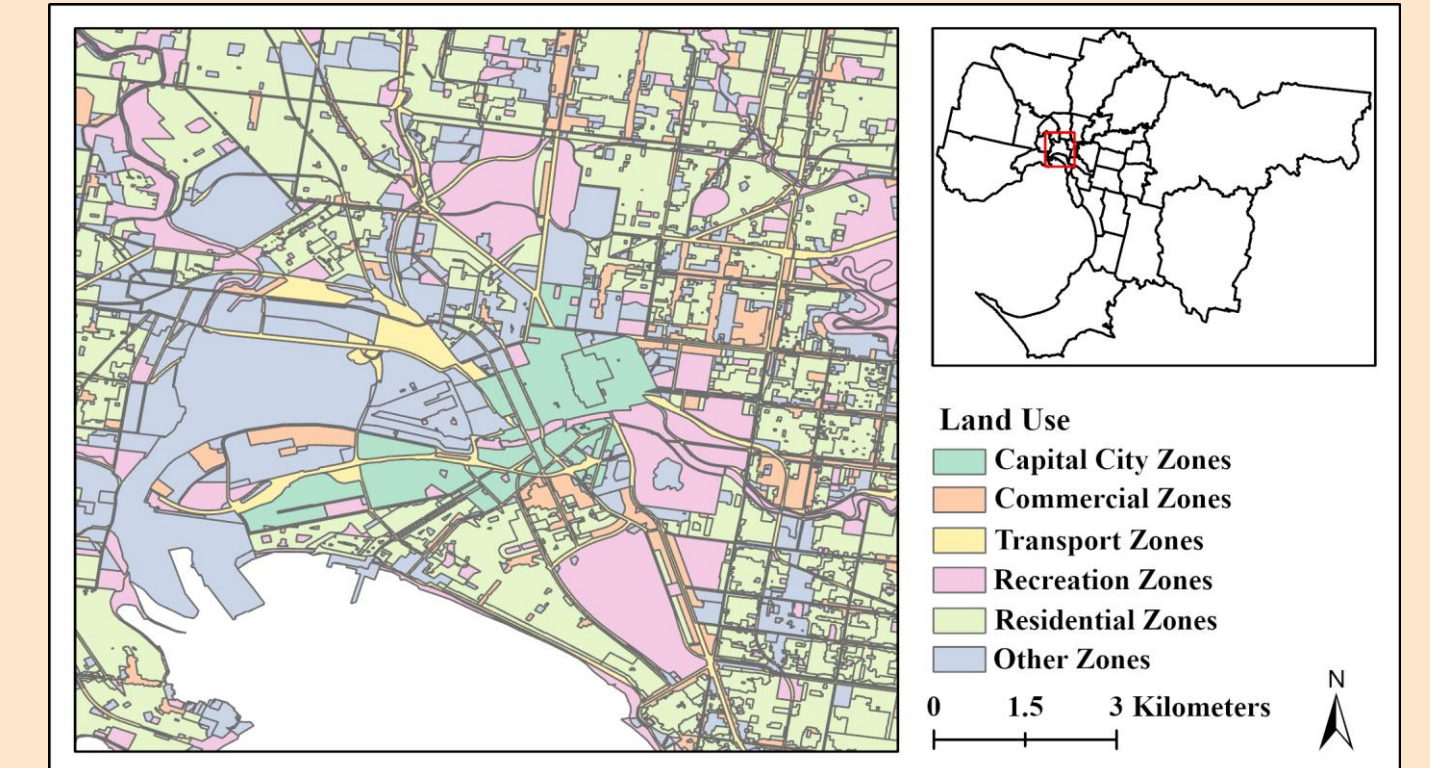


Active transport-related activity locations exhibited clustering patterns during all periods, while the degree of clustering decreased during the pandemic period. This change can be attributed to people's efforts to reduce walking and cycling trips to densely populated areas



Results & Discussion

Active transporters' trips to capital city zones decreased significantly, while their trips to recreation zones such as parks and gardens increased. This shift in travel behavior reflected changes in people's trip motivation during the pandemic period. e.g., a stronger inclination towards leisurely walking and cycling trips



No.	Land Use	Pre-COV number	Pre-COV Ratio (%)	COV number	COV Ratio (%)	Ratio Change
1	Capital City Zones	635	22.38	307	17.49	-4.89
2	Transport Zones	402	14.17	285	16.24	2.07
3	Recreation Zones	647	22.81	417	23.76	0.95
4	Residential Zones	462	16.28	291	16.58	0.30
5	Commercial Zones	252	8.88	162	9.23	0.35
6	Other Zones	439	15.47	293	16.70	1.22

Conclusion & Contribution

- This study makes significant contributions to the field of travel behavior analysis by providing spontaneous data sources that can overcome the challenges of obtaining large amounts of travel data, particularly during the pandemic period.
- This approach also greatly simplifies the retrieval of historical travel records, making retrospective studies more feasible.
- More importantly, the proposed framework offers a solution to the incomplete dataset issue by extracting and supplementing vital activity location information from non-geotagged social media data.
- The integration principles (reasoning rules) developed in this study can serve as a template for location fusion in other travel behavior studies.
- The natural language processing techniques employed in this study, including the BERT-based classifier for tweet classification and the NEM method for content-based location extraction, are promising tools that can be easily transferred to future travel behavior studies to simplify data preparation and location information extraction